

HDOC Day-13 Project

Aim- To write a menu-driven C program using switch-case to perform operations on the digits of a positive integer.

1. Algorithm

1. Start.
2. Enter a positive integer n.
3. Display the menu:
 - 1. Count even and odd digits
 - 2. Find sum of even and odd digits
4. Enter the user's choice.
5. Use switch(choice).
6. If **choice = 1**:
 - Take one digit using $\text{digit} = n \% 10$.
 - Check whether the digit is even or odd.
 - Increase even_count or odd_count.
 - Remove the digit using $n = n / 10$.
 - Repeat until all digits are checked.
7. If **choice = 2**:
 - a. Take one digit at a time.
 - b. Add even digits to sum_even.
 - c. Add odd digits to sum_odd.
 - d. Repeat until all digits are checked.
8. If the choice is not 1 or 2, print **Invalid choice**.
9. Stop.

Test Case	Number	Choice	Expected Output
1	12045	1	Even digits = 3, Odd digits = 2
2	12345	2	Sum even = 6, Sum odd = 9
3	2468	1	Even digits = 4, Odd digits = 0
4	1357	2	Sum even = 0, Sum odd = 16
5	1234	3	Invalid choice

*Write a c program.

```

1 #include <stdio.h>
2
3 int main()
4 {
5     int n, choice, digit;
6     int even_count = 0, odd_count = 0;
7     int sum_even = 0, sum_odd = 0;
8
9     printf("Enter a positive integer: ");
10    scanf("%d", &n);
11
12    printf("\n1. Count even and odd digits");
13    printf("\n2. Sum of even and odd digits");
14
15    printf("\nEnter your choice: ");
16    scanf("%d", &choice);
17
18    switch(choice)
19    {
20        case 1:
21            while(n > 0)
22            {
23                digit = n % 10;
24
25                if(digit % 2 == 0)
26                    even_count++;
27                else
28                    odd_count++;
29
30                n = n / 10;
31            }
32
33            printf("Even digits = %d\n", even_count);
34            printf("Odd digits = %d\n", odd_count);
35            break;
36
37        case 2:
38            while(n > 0)
39            {
40                digit = n % 10;

```

```
21 while(n > 0)
22 {
23     digit = n % 10;
24
25     if(digit % 2 == 0)
26         even_count++;
27     else
28         odd_count++;
29
30     n = n / 10;
31 }
32
33 printf("Even digits = %d\n", even_count);
34 printf("Odd digits = %d\n", odd_count);
35 break;
36
37 case 2:
38     while(n > 0)
39     {
40         digit = n % 10;
41
42         if(digit % 2 == 0)
43             sum_even = sum_even + digit;
44         else
45             sum_odd = sum_odd + digit;
46
47         n = n / 10;
48     }
49
50 printf("Sum of even digits = %d\n", sum_even);
51 printf("Sum of odd digits = %d\n", sum_odd);
52 break;
53
54 default:
55     printf("Invalid choice\n");
56 }
57
58 return 0;
59 }
60 }
```

*Execution, Compilation and test the C program.

```
Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
[priyadarshna@kali:~]$ gcc day13.c -o day13
[priyadarshna@kali:~]$ ./day13
Enter a positive Integer: 12845
1. Count even and odd digits
2. Sum of even and odd digits
Enter your choice: 1
Even digits = 3
Odd digits = 2
[priyadarshna@kali:~]$ ./day13
Enter a positive Integer: 12345
1. Count even and odd digits
2. Sum of even and odd digits
Enter your choice: 2
Sum of even digits = 6
Sum of odd digits = 9
[priyadarshna@kali:~]$
```

